

Chemistry Exam 16 — Chapters 1-9 (Full Syllabus), Detailed Marking Report

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12/12

SECTION A (MCQS)

32/33

SECTION B (SHORT)

20/20

SECTION C (LONG)

64/65

TOTAL, 98.5%

A+

GRADE

Chemistry's 2nd-highest score to date — behind only exam-15's perfect 100%. Section C is flawless (20/20), MCQs are a clean 12/12 (Chemistry's 2nd consecutive perfect MCQ score, after exam-15), and only two small deductions in Section B keep this off a perfect paper.

SECTION A: Multiple Choice Questions (12/12)

#	Question summary	Correct	Seemab	Result	Marks
1	Branch of chemistry protecting water poisoned by soil/dust	A: Environmental chemistry	A	Correct	1
2	Salt whose solubility DECREASES from 20°C to 100°C	D: Ca(OH) ₂	D	Correct	1
3	Isotope symbol: number at upper-left represents	B: Mass number	B	Correct	1
4	Radioactive Cobalt-60 is commonly used to	C: Irradiate/shrink cancer tumours	C	Correct	1
5	Displacement reaction that will occur (F ₂ >Cl ₂ >Br ₂ >I ₂)	B: Br ₂ + 2KI → 2KBr + I ₂	B	Correct	1
6	Correct statement on noble-gas electron configurations	C: He (1s ²) is the exception	C	Correct	1
7	Covalent compound identified from melting points	C: Ethanol	C	Correct	1
8	Molar mass of urea, CO(NH ₂) ₂	A: 60 g/mol	A	Correct	1
9	Na ₂ CX ₃ , formula mass 106 amu → atomic mass of X	C: 16	C	Correct	1
10	Silver decorative objects tarnish due to formation of	B: Silver sulphide	B	Correct	1
11	Branch studying energy changes in reactions	B: Chemical energetics	B	Correct	1
12	N ₂ (g) + 2O ₂ (g) ⇌ 2NO ₂ (g) is produced industrially using	C: An electric spark, no catalyst	C	Correct	1

Section A: 12/12, a completely clean sweep — note this paper had the MCQs written as a plain numbered list on the final scan page (page 9) rather than a bubble panel; all twelve were legible and checked individually against the key. This is Chemistry's 2nd consecutive perfect MCQ score, following exam-15.

SECTION B: Short Questions (32/33)

Q.2(i), MAIN: Define medicinal chemistry and polymer chemistry; one example application of each

2.5/3

Medicinal chemistry: "studies the interaction between drugs and biological targets, and the development of new medicinal targets" (key: agents) — core definition correct. Polymer chemistry: "studies polymers, their types, their behaviour, their structure and uses. For example, nylon, teflon coated bags" — correct with an example given.

−0.5: an example application was given for polymer chemistry (nylon, Teflon-coated bags) but not for medicinal chemistry (the key expects something like "medicines synthesized using knowledge of proteins/enzymes"). The question asks for one example application of EACH branch.

Q.2(ii), MAIN: Differentiate concentrated vs dilute solution, using brine as an example

3/3

"A dilute solution contains a relatively small amount of dissolved solute, while a concentrated solution contains a relatively large amount of solute in the same amount of solvent. Brine is a concentrated solution because it holds a large amount of dissolved NaCl." Matches the answer key exactly.

Q.2(iii), OR: Maximum electrons in s, p, d sub-shells; maximum capacity of the M-shell (n=3)

3/3

s = 2, p = 6, d = 10 (also correctly added f = 14, not required but not penalised). M-shell (n=3) has 3s, 3p, 3d sub-shells, so maximum capacity = 2+6+10 = 18 electrons. Fully correct.

Q.2(iv), OR: Unknown element with valence configuration 3s¹ — group, period, metal/non-metal

3/3

n=3 → Period 3. One valence electron → Group 1. Readily loses this electron, so it is a metal (sodium). Matches the key exactly.

Q.2(v), MAIN: Name FOUR industrial catalysts and the process each is used in

3/3

Iron: Haber process (ammonia synthesis). Platinum/Palladium: catalytic converters. Nickel: manufacture of margarine. Platinum/vanadium pentoxide: Contact process (sulphuric acid). All four correct with matching processes.

Q.2(vi), MAIN: NH₃-BF₃ coordinate bond via lone-pair donation

3/3

"BF₃ has only 3 bonding pairs around boron and is electron-deficient. Ammonia's nitrogen has a lone pair which it donates entirely to boron, forming a coordinate (dative) covalent bond, represented H₃N → BF₃." Textbook-precise, matches the key almost word for word.

Q.2(vii), MAIN: THREE industrial uses of graphite and the structural property responsible for each**2.5/3**

(1) Refractory bricks — chemical inertness, heat stability. (2) Electrodes/anodes in electrolytic furnaces (initially paired with the wrong reason, "carbon brushes in electric motors," which she crossed out and corrected). (3) Moderator rods in nuclear reactors — inertness, plus a bonus 4th use, carbon brushes in electric motors. Four valid graphite uses named in total (only three were required).

–0.5: the structural property/reason was given for uses (1) and (3), but use (2), "electrodes/anodes in electrolytic furnaces," was left without its reason (graphite conducts electricity via the delocalized 4th electron). The question explicitly asks for the property responsible for each use.

Q.2(viii), MAIN: Moles present in 31.6 g of KMnO₄ (K=39, Mn=55, O=16)**3/3**

Molar mass KMnO₄ = 39 + 55 + 4(16) = 158 g/mol. Moles = mass ÷ molar mass = 31.6 / 158 = 0.2 mol. Fully correct, matches the key exactly.

Q.2(ix), OR: Oxidizing/reducing agent in Sn(s) + 2AgNO₃(aq) → Sn(NO₃)₂(aq) + 2Ag(s)**3/3**

Sn: 0 → +2 (oxidized) → Sn is the reducing agent. Ag: +1 → 0 (reduced) → AgNO₃ is the oxidizing agent. Both identified correctly with the oxidation-number justification shown.

Q.2(x), MAIN: Any THREE Key Points about chemical energetics**3/3**

Wrote all five key points from the textbook, well beyond the three required: the study of energy changes in a reaction is chemical energetics; a reaction releasing heat is exothermic (spelled phonetically as "enotermic" both times it appears — see note below) and one absorbing heat is endothermic; ΔH is negative for exothermic and positive for endothermic reactions; bond breaking always absorbs energy, bond forming always releases energy; a catalyst lowers activation energy without changing the overall enthalpy of reaction. All conceptually correct.

Spelling note (no mark lost): "exothermic" is written as "enotermic" on both occurrences. Every property attached to it (releasing heat, ΔH negative) is scientifically correct, so this reads as a phonetic spelling slip rather than a conceptual mix-up — worth a quick flash-card check on the correct spelling.

Q.2(xi), OR: Balance C₂H₆(g) + O₂(g) → CO₂(g) + H₂O(g)**3/3**

2C₂H₆ + 7O₂ → 4CO₂ + 6H₂O. Checked atom-by-atom: C = 4 = 4, H = 12 = 12, O = 14 = 14. Fully correct, matches the key exactly.

Section B: 32/33. All 11 required questions were attempted and every sub-part completed — no early stops anywhere on the paper, continuing the positive trend since this used to be Seemab's most common source of lost marks. The two small deductions are both "one part of a paired requirement given, the other omitted" (a missing example, a missing justification), not incomplete answers.

SECTION C: Long Questions (20/20)**Q.3(i), MAIN: (a) Isotope B's mass from reverse RAM calculation [3] (b) Anion formation, S and Cl [2]****5/5**

(a) Total = RAM × 100 = 35.5 × 100 = 3550. Contribution of A = 75.77 × 35 = 2651.95. Contribution of B = 3550 – 2651.95 = 898.05. Isotopic mass of B = 898.05 ÷ 24.23 ≈ 37.06 amu. Matches the key exactly. (b) S (electron config 1s²2s²2p⁶3s²3p⁴) gains 2 electrons → forms the –2 (S^{2–}) anion. Cl (1s²2s²2p⁶3s²3p⁵) gains 1 electron → forms the –1 (Cl[–]) anion. Both electrons-gained and resulting charges correctly stated for both elements.

Q.3(ii), MAIN: (a) Empirical/molecular formula, 85.7% C / 14.3% H, M=56 g/mol [3] (b) Balance P₄(s) + O₂(g) → P₂O₅(s) [2]**5/5**

(a) C: 85.7/12 = 7.14. H: 14.3/1 = 14.3. Ratio 1:2 → empirical formula CH₂ (mass 14). n = 56/14 = 4 → molecular formula C₄H₈. Matches the key exactly. (b) P₄ + 5O₂ → 2P₂O₅. Checked: P = 4 = 4, O = 10 = 10. Correctly balanced.

Q.3(iii), MAIN: (a) Oxidation states of Cr in Cr₂(SO₄)₃ and N in NaNO₃ [3] (b) Oxidizing/reducing agent in 2Al(s) + 3CuSO₄(aq) → Al₂(SO₄)₃(aq) + 3Cu(s) [2]**5/5**

(a) Cr₂(SO₄)₃: 2x + 3(–2) = 0 → 2x – 6 = 0 → x = +3. NaNO₃: 1 + x + 3(–2) = 0 → x – 5 = 0 → x = +5. Both correct with working shown. (b) Al: 0 → +3 (oxidized) → reducing agent = Al. Cu: +2 → 0 (reduced) → oxidizing agent = CuSO₄. Both correctly identified.

Q.3(iv), MAIN: (a) Diamond vs graphite structure [3] (b) Uses of diamond and graphite [2]**5/5**

(a) "In diamond, each C bonds to 4 others in a rigid 3-D tetrahedral network with no free electrons → extremely hard, electrical insulator. In graphite, each C bonds to only 3 others in 2-D hexagonal layers; the 4th delocalized electron moves freely within each layer (conducts electricity), and weak van der Waals forces between layers let them slide → soft, slippery, conducting." Matches the key closely and completely. (b) Diamond: cutting/drilling tools (hardness); medical tools e.g. cataract surgery (hardness/precision). Graphite: electrodes in furnaces (conducts + heat resistant); pencils/lubricant (soft, layered). All four uses given with matching reasons.

Section C: 20/20. All four long questions — isotope arithmetic, empirical/molecular formula, oxidation-state derivations, and the diamond/graphite structure-property comparison — completed with full working and correct final answers throughout.

Deduction Log (reconciles to headline total)

Section	Marks available	Marks lost	Reason	Marks awarded
A: MCQs	12	0	None	12
B: Short questions (11)	33	1.0	Q2(i) –0.5 (missing medicinal-chemistry example); Q2(vii) –0.5 (missing property/reason for one graphite use)	32
C: Long questions (4)	20	0	None	20
Total	65	1.0	Two half-mark deductions, both Section B	64

Strengths

- Perfect MCQ panel, 12/12 — Chemistry's 2nd consecutive perfect MCQ score, after exam-15
- Section C flawless, 20/20: isotope reverse-calculation,

Areas to Revise

- Q2(i) –0.5:** gave an example application for polymer chemistry but not for medicinal chemistry. When a question asks for "one example application of EACH," make sure both halves get one.

empirical/molecular formula, balancing $\text{P}_4 + 5\text{O}_2 \rightarrow 2\text{P}_2\text{O}_5$, both oxidation-state derivations, redox agent identification, and the diamond/graphite comparison all correct with full working

- Every one of the 11 Section B questions attempted with every sub-part completed — no early stops, continuing the positive trend on Seemab's most historically common weakness
- Stoichiometry precise throughout: KMnO_4 molar mass (158 g/mol) $\rightarrow$ 0.2 mol, and C_2H_6 combustion balanced and atom-checked
- Sn/AgNO_3 redox roles and the $\text{NH}_3\text{-BF}_3$ coordinate-bond mechanism both stated letter-perfect against the key

- **Q2(vii) –0.5:** named three (plus a bonus fourth) valid graphite uses, but the structural-property reason was only written for two of them. Every named use needs its own reason attached.
- **Spelling watch (no marks lost):** "exothermic" written as "enotermic" twice in Q2(x). The science was correct both times — just double-check the spelling.

Comparison vs Previous Chemistry Attempts

Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
12 Apr 2026	exam-01-ch1-2	12/12	31/33	21/20*	64/65	98.5%	A+
17 Apr 2026	exam-02-ch3-5	12/12	31/33	16.5/20	59.5/65	91.5%	A+
12 May 2026	exam-03-ch6	12/12	30.5/33	15/20	57.5/65	88.5%	A+
12 Jun 2026	exam-04-ch7	12/12	30/33	18/20	60/65	92.3%	A+
24 Jun 2026	exam-12-ch1-9	12/12	31.5/33	19/20	62.5/65	96.2%	A+
30 Jun 2026	exam-13-ch1-9	12/12	31.5/33	17/20	60.5/65	93.1%	A+
6 Jul 2026	exam-14-ch1-9	11/12	31.5/33	17.5/20	60.0/65	92.3%	A+
14 Jul 2026	exam-15-ch1-9	12/12	33/33	20/20	65/65	100%	A+
22 Jul 2026	exam-16-ch1-9 (this exam)	12/12	32/33	20/20	64/65	98.5%	A+

* Chemistry exam-01 Section C shows 21/20 due to a historical bonus mark in the original marking.

exam-16 (64/65, 98.5%) is Chemistry's 2nd-best score to date, just behind exam-15's perfect 100%, and its 2nd-best full-syllabus Ch1-9 paper (exam-12: 96.2%, exam-13: 93.1%, exam-14: 92.3%, exam-15: 100%, exam-16: 98.5%). It is also the first time Chemistry has posted two consecutive perfect Section C results (exam-15 and exam-16, both 20/20) and two consecutive perfect MCQ scores. Only two small, isolated half-mark deductions in Section B (a missing example, a missing justification) separate this paper from a second straight perfect score.